Exercise Spivak §1 p15

Prove that if $x, y \neq 0$, then

$$x^2 + xy + y^2 > 0 \quad (*)$$

Pf Note that

$$x^3 - y^3 = (x-y)(x^2+xy+y^2)$$

Case 1 $x = y$

Then $(*)$ is $x^2 + x^2 + x^2 > 0$

Case 2 $x > y$

Then $x^3 > y^3$

This is equivalent to

$$x^3 - y^3 > 0.$$

Then $\underbrace{(x-y)}_{}(x^2+xy+y^2) > 0$

* Assumed $x > y$
So $x-y > 0$

So $x^2+xy+y^2 > 0$.

II. Numbers of Various Sorts

$\mathbb{N} = \{1, 2, 3, \ldots\}$

$\Rightarrow$ Mathematical induction

(If you want to define $\mathbb{N}$ from a small set of axioms induction is one of them $\Rightarrow$ Peano Arithmetic)

If $P(k)$ is a statement that is either true or false for each value of $k \in \mathbb{N}$.

and if: ① $P(1)$ is true

② If $P(k)$ is true, then $P(k+1)$ is true.

$\Rightarrow$ Then, $P(k)$ is true for every $k \in \mathbb{N}$.

In $\mathbb{N}$, this is taken as an axiom.

Ex. Show that $P(k)$ is true for all k.

$$P(k) = 1 + \cdots + k = \frac{k(k+1)}{2}.$$

Pf. Taking induction for granted.

need to verify ① and ② to show that $P(k)$ is true $\forall k$.

① $P(1)$: $1 = \frac{1(2)}{2} = 1$. $\boxed{\text{True}}$

②. Assume $P(k)$ is true

want $P(k+1)$ is true

Add $k+1$ to both sides

$$(1+\cdots+k)+(k+1) = \frac{k(k+1)}{2}+k+1$$

$$\overset{(pa)}{=} (k+1)\left(\frac{k}{2}+1\right)$$

$$= (k+1)\left(\frac{k+2}{2}\right)$$

By induction, $P(k)$ is true $\forall k$.

$\blacksquare$

Well-ordering Principle

if $A \subseteq \mathbb{N}$, $A \neq \emptyset$
then A has the least element

This is equivilant to induction.

Pf (Induction $\Rightarrow$ WOP)

Suppose for a contradiction that $\exists A \subseteq \mathbb{N},\ A \neq \emptyset$ without a least element.

Define $P(k)$: $1, \cdots, k \notin A$

Apply induction to P.

① Is $p(1)$ true?

Yes, if $1 \in A$, then 1 is its least element.

② Assume $p(k)$ is true. Show $p(k+1)$ true

$$\left.\begin{array}{l} \Rightarrow \text{At least one of } 1, 2, \cdots, k+1 \in A \\[2em] \text{But } p(k) \text{ says } 1, \cdots, k \in A \end{array}\right\} \ k+1 \in A$$

In this case, $k+1$ is the least element of A

Since A was assumed to have a least element, this is a contradiction.

∎

(WOP $\Rightarrow$ Induction)

Assume every $\phi \neq A \subseteq \mathbb{N}$ has a least element

Let $A = \{ k \in \mathbb{N} : P(k) \text{ is false} \}$

If A were not empty, by WOP, it has least element $l \in A \subseteq \mathbb{N}$.

($p(l)$ is false)

So. $l \neq 1$ since $p(l)$ is false.

$P(l-1)$ is true b/c l is the least elt in A.

By induction. $p(l-1)$ true $\Rightarrow$ $p(l)$ true

So $l \notin A$. Contradiction $\Rightarrow$ $A \neq 0$

$\blacksquare$

Strong Induction

$P(k)$ property that is true or false for each $k \in \mathbb{N}$.

Assume ① $P(1)$ is true.

② If $P(1), \dots, P(k-1)$ all true $\Longrightarrow P(k)$ is true.

Then, $P(k)$ true for all $k \in \mathbb{N}$.